Proceedings of the 33rd Chinese Control Conference
July 28-30, 2014, Nanjing, China

The Offline Feature Extraction of Four-class Motor Imagery EEG Based on ICA and Wavelet-CSP

BAI Xiaoping[1], WANG Xiangzhou[1], ZHENG Shuhua[1*], YU Mingxin[1]

1. School of Automation, Beijing Institution of Technology, Beijing 100081, P.R.China
E-mail: 13426116792@139.com

Abstract: The signal processing of electroencephalogram (EEG) is the key technology in a brain-computer interface (BCI) system. A widely used method is to purify the raw EEG with an 8-30Hz band-pass filter and extract features by common spatial patterns (CSP). However its results for BCI Competition IV are not very satisfactory. To improve the classification success rate, this paper proposed a novel Wavelet-CSP with ICA-filter method. For the data sets from BCI Competition IV, the features of the four-class motor imagery were trained and tested using the Support Vector Machines (SVM). The experimental results showed that the proposed method had a higher average kappa coefficient of 0.68 than 0.52 of the general method.

Key Words: Brain-computer-interface (BCI), electroencephalogram (EEG), ICA, Wavelet-CSP, SVM

1 Introduction

The brain-computer interface (BCI) is a non-muscular communication channel for exchanging messages and commands between the human brain and the computer[1]. By means of BCI, people suffered from physical disability are able to control their wheelchairs or prostheses. The essence of a BCI is to identify the features of special EEG signals and then convert them into commands. Based on different EEG signals (μ and β rhythms, slow cortical potentials, event-related P300, visual evoked potentials and so on), different BCI systems were born. Among those BCI systems, the motor imagery-based BCI stands out for its advanced approach without the help of body movement as well as the low requirement for auxiliary equipment. Therefore, it has been more accepted and widely researched.

It has been shown in [2] that motor imagery could cause the energy of μ and β rhythms (around 10Hz and 20Hz) increased or decreased in the related cortical areas, which has been labeled event-related synchronization (ERS) or desynchronization (ERD). Several factors reveal that left or right hand motor imagery could cause the contralateral hemisphere ERD and the ipsilateral hemisphere ERS. Foot motor imagery may not only result in the ERD closed to the foot area (Cz), but also can cause the ERS over the hand area (C3, C4). The obvious different between foot and tongue motor imagery is the mid-central (Cz) ERD with foot motor imagery however mid-central (Cz) ERS with tongue motor imagery[3][4]. Therefore, by detecting the energy changes of the related areas around 10Hz and 20Hz, different motor imagery patterns could be found out. Depending on this method, Pfurtscheller and his team have developed the first motor imagery-based BCI[5].

However, over the past few years, the researchers mainly focused on classifying the left and right hand motor imagery[6], which has only two categories and significant limitations when applied in reality. This paper will investigate a BCI based on four-class motor imagery including left hand, right hand, foot and tongue motor imagery using the multichannel data from Data sets 2a of BCI Competition IV[7].

The key parts of a BCI are signal preprocessing and feature extraction. As we know, EEG signals are sensitive to the electrical interference and artifacts, such as the 50Hz frequency interference, the circuit noise, Electrooculogram (EOG), Electrocardiograph (ECG) and Electromyography (EMG). Therefore, removing the noise is necessary before feature extraction. There are a variety of options, for example, the band-pass filters, the Laplacian derivation, the Common Average Reference spatial filters and Independent Components Analysis (ICA)[8], which can be used to improve signal-to-noise ratios of the EEG signals. Compared with other methods, ICA is more powerful to separate the useful signals from the multichannel mixed-signals just like the raw EEG[9]. Currently, the Common Spatial Patten (CSP) with optimal spatial filters has been proven to be an effective feature extraction algorithm for the two-class EEG[10]. For discrimination of the four-class problem, One-Versus-One (OVO) CSP and One-Versus-the-Rest (OVR)[11] CSP have been proposed. Several improved CSP also appeared, such as Sub-band Common Spatial Patten (SBCSP)[12] and Filter Band Common Spatial Patten (FBCSP)[13]. The novel idea of the SBCSP and FBCSP is to decompose the raw EEG into multiple frequency bands and employ the CSP on each band to select the most discriminative features. However, the decomposition to narrow frequency bands may damage the raw signal, and the classification results may be also affected. This paper proposed a novel method that combined the Wavelet Analysis and conventional CSP, named Wavelet-CSP, to solve a multi-class issue by simplifying it into several binary problems. According to ERD/ERS, the difference between two kinds of motor imagery is higher in the frequency domain than that in the time domain. Consequently, the Wavelet-CSP directly used the wavelet coefficients of the feature band instead of the sampling values of the signals as the input of the CSP to extract the features. In addition, since CSP is sensitive to noise, the classification success rate will debase along with the deterioration of the signal quality. Thus, this paper designed an ICA-filter to remove noise before the Wavelet-CSP. Finally, the support vector machine (SVM)[14] was used to design a classifier that differed to OVO and OVR to distinguish the four-class motor imagery.

2 Methods

2.1 Experiments and Data Description

The data sets 2a applied in this paper were provided by the Institute for Knowledge Discovery, Graz University of Technology[15]. EEG signals of nine subjects were recorded during the cure-based motor imagery tasks, namely the movement imagination of the left hand, right hand, foot and tongue.

Four stages were included in a trial with 7.5 seconds. It is shown in Fig. 1. First is the preparation stage. A fixation cross appeared in front of the subjects and lasted for 2s. Next is the prompt stage. An arrow-shaped cue appeared to indicate which movement to be imagined and stayed for 1.25s. The third is the stage of movement imagination. The subjects were asked to execute the motor imagery tasks until the fixation cross disappeared at t=6s. Last, a resting stage is followed for 1.5s.

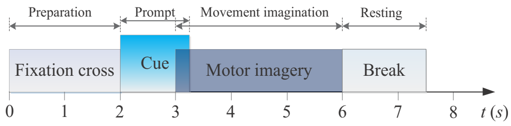

Fig. 1: The four stages of a trial

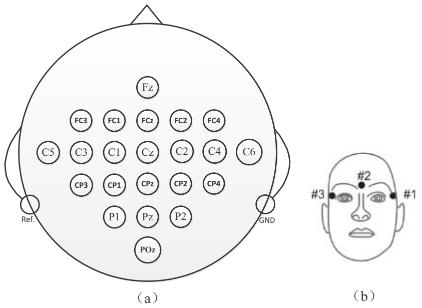

Fig. 2: (a) The locations of the EEG electrodes. (b) The locations of the EOG channels

The EEG data was recorded by twenty-two Ag/AgCl electrodes and the locations of the electrodes are shown in Fig.2(a). Another three EOG channels were provided as Fig.2(b). The EEG signals were sampled at 250 Hz and band-pass filtered between 0.5 Hz and 100 Hz.

2.2 Artifacts Removing based on ICA

The sampled signals in each channel are always a linear mixture of irrelevant motor imagery signals and artifacts, such as EOG, ECG and EMG. Thus, ICA can be employed to remove the artifacts and reconstruct the motor imagery signals for the multichannel EEG. The basic model of ICA is described as Fig. 3[16].

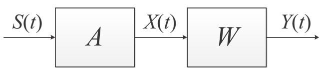

Fig. 3: The basic model of ICA

where $S(t)=[s_1(t), s_2(t), \cdots, s_m(t)]^T$ denote m source signals. $X(t)=[x_1(t), x_2(t), \cdots, x_n(t)]^T$ denote n-channel scalp EEG signals. A is the mixing matrix and W is the un-mixing matrix. The separated independent components (ICs) are $Y(t)=[y_1(t), y_2(t), \cdots, y_m(t)]^T$.

$$Y(t) = WX(t) \tag{1}$$

In this paper, negative entropy-based FastICA algorithm was chosen. It mainly includes two parts: data whitening and separation vectors estimating. The steps of FastICA are introduced in [17]. The basic idea of FastICA is to find out a projection direction w^T to maximize local non-Gaussian of $w^T Z$ on the condition of $\| w \|_2 = 1$, where Z is the whitening of X . The iterative formula to acquire w^T is followed:

$$w = E\left\{zg\left(w^T z\right)\right\} - E\left\{g'\left(w^T z\right)\right\}w \tag{2}$$

where g is generally defined by the followed three functions:

$$g_1(u) = \tanh(u), g_2(u) = u\exp\left(-\frac{u^2}{2}\right), g_3(u) = u^3 \tag{3}$$

After ICA, an un-mixing matrix W and m ICs are obtained. With the decomposition of the raw EEG, an ICA-filter is designed as Fig. 4.

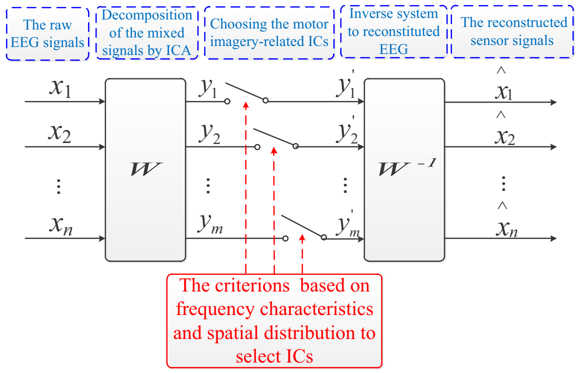

Fig. 4: The process of the ICA-filter

The Fig. 4 shows that if the motor imagery-related ICs $Y'(t) = [y_1'(t), y_2'(t), \cdots, y_m'(t)]^T$ are figured out, the reconstructed EEG $\hat{X}(t) = [\hat{x}_1(t), \hat{x}_2(t), \cdots, \hat{x}_n(t)]^T$ can be calculated by the inverse system as follow:

$$\hat{X} = W^{-1}Y' \tag{4}$$

Hence, identification of the motor imagery-related ICs from $Y(t)$ in the process of ICA-filter is the most crucial step within the whole procedure. Two criterions were proposed in this paper. First, the frequency characteristic must be taken into consideration[18]. It is confirmed that the frequency of the motor imagery-related signals are mainly in μ and β rhythms around 10Hz and 20Hz. Therefore, the matching of the frequency characteristics can be regarded as evidences of the motor-related ICs. Second, the spatial distribution needs to be verified by cross-correlation. This paper checked the correlation between the ICs and the raw EEG of C3, Cz, C4 in the related cortical areas. The correlation coefficient should be greater if they were motor-related ICs than the ICs of artifacts. Through the above two methods, the motor-related ICs should be figured out. At last, this paper reserved the useful ICs and set the rest to zero. Thus $Y'(t) = [y_1'(t), y_2'(t), \cdots, y_m'(t)]^T$ was acquired as

$$y_i' = \begin{cases} y_i & y_i \in \text{motor imagery ICs} \\ [\underbrace{0\ 0 \cdots 0}_{N}]^T & y_i \in \text{artifact ICs} \end{cases} \tag{5}$$

where N is the length of y_i

Fig. 5 shows the performance of the ICA-filter and a contrast in the frequency domain between the 8~30Hz band-pass filtered EEG and the ICA-filtered EEG. The data was selected from 3s to 6s of a trial that contaminated by the EOG seriously. From Fig. 5(a) and Fig. 5(b), it can be noticed that artifacts are largely eliminated and the reconstructed EEGs become smooth and clean. Fig. 5(c) demonstrates that the ICA-filter has a better frequency characteristic than the band-pass filter and is highly effective to retain the main frequency around the 10Hz and remove the useless interferential frequency.

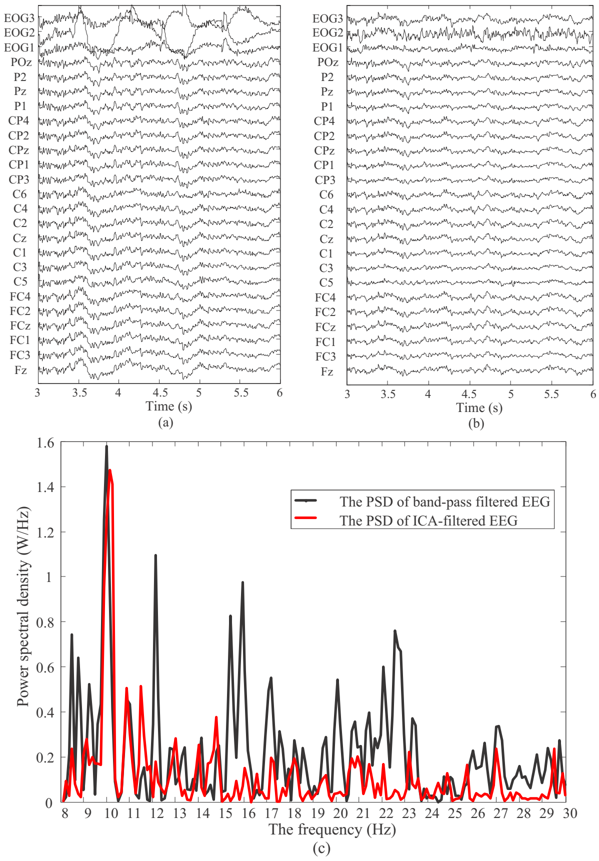

Fig. 5: (a) The raw EEG signals; (b) The ICA-filtered EEG signals; (c) The contrast of the power spectral density(PSD) of C3 between the band-pass filtered EEG and the ICA-filtered EEG in 8~30Hz

2.3 Feature Extraction using Wavelet-CSP

(a) Common Spatial Patterns (CSP)

The method of CSP is to decompose the raw EEG signals into spatial patterns that maximize the differences between two classes. It has successfully been applied to distinguish the movement imagination of left and right hands[19]. The specific process of this algorithm is expressed as follows:

(1) Calculate the normalized spatial covariance of EEG signals. The two kinds of raw EEG data can be expressed by the $N \times T$ matrixes: X_1, X_2, where N is the number of channels, T is the number of samples per channel. The normalized spatial covariance of EEG can be obtained by

$$R_i = \frac{X_i X_i^T}{\mathrm{trace}(X_i X_i^T)}, i = 1,2 \tag{6}$$

where trace(x) is the sum of the diagonal elements of matrix x.

(2) Employ the eigenvalue decomposition on the sum of the spatial covariance. It can be giving as

$$R = R_1 + R_2 = U_0 \Lambda U_0^T \tag{7}$$

where U_0 is the matrix of eigenvectors and Λ is the diagonal matrix of eigenvalues.

(3) Obtain the whitening matrix and whiten the covariance. The whitening transformation is

$$W = \Lambda^{-1/2} U_0^T \tag{8}$$

The covariance can be whitened by

$$S_1 = W R_1 W^T \tag{9}$$

and
$$S_2 = W R_2 W^T \tag{10}$$

It can be proved that S_1 and S_2 share common eigenvectors[20], that is

if
$$S_1 = U \Lambda_1 U^T \tag{11}$$

then
$$S_2 = U \Lambda_2 U^T \tag{12}$$

and
$$\Lambda_1 + \Lambda_2 = I \tag{13}$$

where I is the identity matrix. Since the sum of two corresponding eigenvalues is always one, the eigenvector with largest eigenvalue for S_1 has the smallest eigenvalue for S_2 and vice versa. Thus, this property makes U useful for classification of two populations.

(4) Acquire the projection matrix and decompose a trial. Here, the $2m$ eigenvectors corresponding to the m largest eigenvalues in Λ_1 and Λ_2 respectively were chosen sequentially to form a new matrix U' that are optimal for discriminating two classes. The projection matrix can be simplified by

$$SF = (U')^T W \tag{14}$$

The decomposition of a trial X_i is given as

$$Z_i = SF \cdot X_i \tag{15}$$

The variance of Z_i can be used as features to be classified.

(b) The Wavelet-CSP algorithm

Because the differences between the time-domain signals are very small during imagination, it is hard to find a global optimal projection matrix through the sampling values. However, experiments show that the differences of the frequency values in μ and β rhythms are more apparent according to ERD/ERS and the optimal projection matrix based on the frequency values can be easier to train. Thus, a novel Wavelet-CSP algorithm that uses the wavelet coefficients as the input of the CSP instead of the sampling values was employed in this paper.

It is well known that the signal $f(t)$ can be decomposed by Wavelet Transform[21] as

$$f(t) = A_L + \sum_{j=1}^{L} D_j = A_L + D_L + D_{L-1} + \cdots + D_1 \tag{16}$$

where L is the decomposition layer, A_L is the low-pass approximation component, D_j is the detail component under different scales. If the sample frequency of $f(t)$ is f_s, the sub-frequency bands of all components will be $[0, \frac{f_s}{2^{L+1}}]$, $[\frac{f_s}{2^{L+1}}, \frac{f_s}{2^L}], [\frac{f_s}{2^L}, \frac{f_s}{2^{L-1}}], \cdots, [\frac{f_s}{2^2}, \frac{f_s}{2}]$, successively. The corresponding coefficients of approximation component and each detail component can be denoted by $cA_L, cD_L, cD_{L-1}, \cdots cD_1$.

In order to obtain the wavelet coefficients of 8~30 Hz for the signals with 250Hz sampling frequency, a four-layer Wavelet Transform was applied. According to the rules of the decomposition, the sub-frequency bands should be [0,7.8125], [7.8125,15.625], [15.625,31.25], [31.25,62.5], [62.5,125], theoretically. The coefficients should be cA_4, $cD_4, cD_3, \cdots, cD_1$ corresponding to the sub-frequency bands. Because the frequency bands of [7.8125,15.625] and [15.625,31.25] include the μ and β rhythms, the related wavelet coefficients $[cD_4, cD_3]$ should reflect their frequency characteristics. The wavelet coefficients $[cD_4, cD_3]$ for C3 and C4 during imagination of left hand are shown in Fig. 6. It can be found that large difference appears corresponding to ERD/ERS.

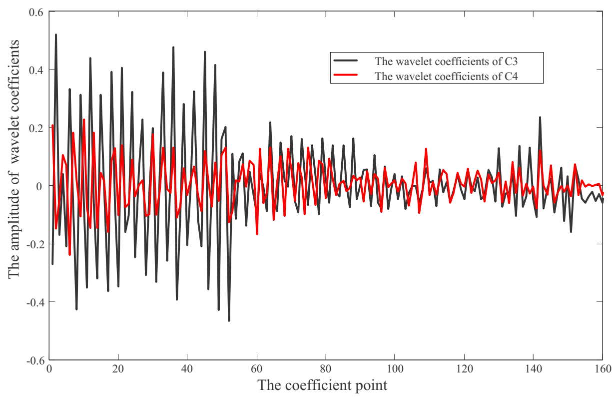

Fig. 6: The contrast of the wavelet coefficients for C3 and C4 when imagination of left hand

Consequently, each channel of the sampling values was replaced by the sequence of $[cD_4, cD_3]$ as the input of CSP. The $N \times T$ matrix X_i (i=1, 2) of the EEG signals was transformed to be a $N \times L$ coefficient matrix C_i (i=1, 2). L is length of the wavelet coefficients of a single channel. A new projection matrix SF' was obtained to decompose the C_i as

$$Z_i' = SF' \cdot C_i, \; i=1, 2 \tag{17}$$

where Z_1' and Z_2' represent the decomposed matrix of the two classes.

2.4 Classification

After decomposition of C_i by the projection matrix SF', a $2m \times L$ matrix Z_i' is obtained. The variance of each row of Z_i', denoted by $\mathrm{VAR}_{ij}\,(j = 1, 2, \cdots, 2m)$, is used as feature for classification[22]. The normalization of the features is

$$f_{i,j} = \frac{\mathrm{VAR}_{ij}}{\sum_{j=1}^{2m} \mathrm{VAR}_{ij}}, i = 1, 2 \tag{18}$$

Thus, the features of the two classes are obtained by

$$\begin{cases} f_1 = [f_{1,1}, f_{1,2}, \cdots, f_{1,2m}]^T \\ f_2 = [f_{2,1}, f_{2,2}, \cdots, f_{2,2m}]^T \end{cases} \tag{19}$$

where f_1 and f_2 represent the features of two types of imagination. The feature vector $[f_1, f_2]$ is used to train the classifier SVM[23].

In this paper, the CSP was used to solve a four-class problem. Therefore, the four-class issue was decomposed to several binary problems that each of them could be classified based on different phenomenon of ERD/EDR. The rules of ERD/ERS around the channels of C3, Cz and C4 during the four-class motor imagery are presented as Table 1.

Table 1: ERD/ERS around the Channels of C3, Cz and C4

Channel	Left hand	Right hand	Foot	Tongue
C3	ERS	ERD	ERS	ERS
Cz			ERD	ERS
C4	ERD	ERS	ERS	ERS

Table 1 shows that ERD appears around the channels of C3 or C4 during the motor imagination of hands, but does not occur when motor imagination of foot or tongue. Thus, a projection matrix SF_1 and SVM1 could be trained to distinguish these two kinds.

Besides, there are different areas of ERD/ERS around C3 and C4, a projection matrix SF_2 and SVM2 could be trained to classify the left and right hand motor imagery.

Last, due to the adverse ERD/ERS around Cz, a projection matrix SF_3 and SVM3 could be trained to classify the motor imagery of foot and tongue.

Based on the above rules, the structure of the classifier was designed as Fig. 7.

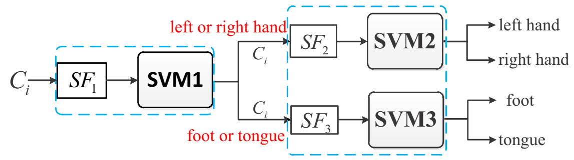

Fig. 7: The structure of the classifier

3 Results

The data sets consists of EEG data from 9 subjects and 576 groups of data (144 for each of the four possible classes) per person, half for training and other half for testing. In this paper, 18 channels around the C3, Cz, C4 in the motor somatosensory area were chosen from each data set and the algorithm was applied on these channels.

3.1 Selection of the Best Time Segment

In order to evaluate the influence of the different time segments on the classification success rate, a 3s sliding time window with a step of 0.5s was applied to the 7.5s raw data and cut raw data into 10 pieces. The average classification results of four conditions in different segments from subject1 are shown in Fig. 8. It shows different time segments bring out different results. At the beginning of a trial, the accuracy is low. When the starting point of the time segment overs 2s, the accuracy becomes higher, because the prompt arrow appears at t=2s. However the accuracy drops down at the end of a trial. The best time segment for subject1 is 2.5~5.5s and the accuracy can reach to 85%~90%. Thus, a suitable time segment is vital for the classification success rate because the suitable time for the short-lasting ERD and ERS varies with each individual[2].

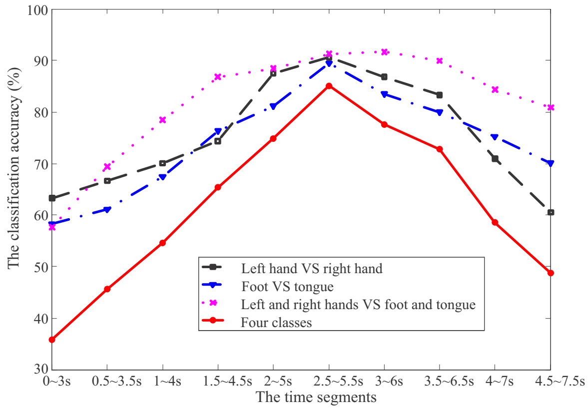

Fig. 8: The average classification results of the following four conditions: left hand VS right hand, foot VS tongue, left and right hands VS foot and tongue, four classes

3.2 The Comparison of Classification Results

The data with suitable time segment was chosen to train the projection matrix and SVM through the 288 groups of training data for each person. Then, the training parameters were applied to another 288 groups of testing data to obtain the results. The procedure is shown in Fig. 9.

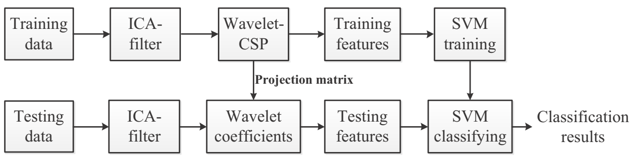

Fig. 9: The procedure for classification of four-task EEG

The results of the proposed method are compared with the results of the conventional CSP based on the 8-30Hz IIR band-pass filters from BCI Competition IV[24]. In this competition, a kappa coefficient k was used as a measure of distinctiveness. The values of k of the two methods are shown in Table 2. The proposed method produces a higher average k value of 0.68 than 0.52 of conventional CSP with a band-pass filter. This means that the proposed method has a higher classification success rate.

Table 2: The Kappa Coefficient Values of Different Subjects

The subjects	ICA+Wavelet_CSP	IIR +CSP
S1	0.75	0.69
S2	0.61	0.34
S3	0.80	0.71
S4	0.63	0.44
S5	0.57	0.16
S6	0.52	0.21
S7	0.77	0.66
S8	0.74	0.73
S9	0.72	0.69
Mean	0.68	0.52

In addition，it can be noticed that the classification success rate is more stable than the conventional method whose results range from 0.16 to 0.73. The relatively small fluctuation indicates that the proposed methods are more adaptable. The main reason for this phenomenon is that conventional CSP is sensitive to noise. The classification success rate will fluctuate along with the change of the signal quality. Researches show that the artifacts and the motor imagery signals have frequency overlap. An 8-30Hz IIR band-pass filter not only cannot remove most influence of artifacts but also may damage the motor imagery signals. However, the ICA method can retain the integral of the motor-related signals better through separating the motor imagery signals from the raw EEG to remove the artifacts. Therefore, even if the quality of raw EEG is poor, the proposed methods can still maintain good classification results.

4 Conclusion

From the study, it can be concluded that on one hand, the ICA-filter is better to remove the artifacts in EEG than the band-pass filter; on the other hand, the global projection matrix of CSP is easier to be trained by the wavelet coefficients than by the sampling values. Therefore, the results gained by the proposed feature extraction approach called Wavelet-CSP based on an ICA-filter had a higher and more stable classification success rate.

References

[1] J.R. Wolpaw, N. Birbaumer, D.J. McFarland, G. Pfurtscheller, T.M. Vaughan, Brain–computer interfaces for communication and control, *Clinical neurophysiology*, 2002, 113(6): 767-791.

[2] G. Pfurtscheller, Neuper C, Motor imagery activates primary sensorimotor area in humans, *Neuroscience letters*, 1997, 239(2): 65-68

[3] G. Pfurtscheller, Ch. Neuper, C. Andrew, G. Edlinger, "Foot and hand area mu rhythms, *International Journal of Psychophysiology*, 1997, 26(1-3): 121-135.

[4] G. Pfurtscheller, C. Brunner, A. Schlögl, F.H. Lopes da Silva, Mu rhythm (de) synchronization and EEG single-trial classifycation of different motor imagery tasks, *NeuroImage*, 2006, 31(1): 153-159.

[5] G. Pfurtscheller, C. Neuper, D .Flotzinger, M. Pregenzer, EEG-based discrimination between imagination of right and left hand movement, *Electroencephalography and clinical Neurophysiology*, 1997, 103(6): 642-651.

[6] G. Pfurtscheller, C. Neuper, A. Schlogl K. Lugger, Separability of EEG signals recorded during right and left motor imagery using adaptive autoregressive parameters, *Rehabilitation Engineering*, *IEEE Transactions on*, 1998, 6(3): 316-325.

[7] G. Pfurtscheller, data sets of BCI Competition IV. Available http://www.bbci.de/competition/iv/#datasets

[8] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, T. M. Vaughan. Brain–computer interfaces for communication and control, *Clinical neurophysiology*, 2002, 113(6): 767-791.

[9] C. Brunner, M. Naeem, R. Leeb, G. Bernhard, G. Pfurtscheller, Spatial filtering and selection of optimized components in four class motor imagery EEG data using independent components analysis, *Pattern Recognition Letters*, 2007, 28(8): 957-964.

[10] J. Müller-Gerking, G. Pfurtscheller, H. Flyvbjerg, Designing optimal spatial filters for single-trial EEG classification in a movement task, *Clinical neurophysiology*, 1999, 110(5): 787-798.

[11] W. Wu, X. Gao, S. Gao, One-versus-the-rest (OVR) algorithm: An extension of common spatial patterns (CSP) algorithm to multi-class case, *Engineering in Medicine and Biology Society, 2005. IEEE-EMBS 2005. 27th Annual International Conference of the. IEEE*, 2006: 2387-2390.

[12] Q. Novi, C. Guan, TH. Dat, P. Xue, Sub-band common spatial pattern (SBCSP) for brain-computer interface, *Neural Engineering, 2007. CNE'07. 3rd International IEEE/EMBS Conference on. IEEE*, 2007: 204-207.

[13] K. K. Ang, Z. Y. Chin, H. Zhang, C. Guan, Filter bank common spatial pattern (FBCSP) in brain-computer interface, *Neural Networks, 2008. IJCNN 2008. (IEEE World Congress on Computational Intelligence)*, *IEEE International Joint Conference on*. *IEEE*, 2008: 2390-2397.

[14] F. Lotte, M. Congedo, A. Lécuyer, F. Lamarche, B. Arnaldi, A review of classification algorithms for EEG-based brain–computer interfaces, *Journal of neural engineering*, 2007, 4.

[15] C. Brunner, R. Leeb, G. Müller-Putz, A. Schlögl, G. Pfurtscheller, BCI Competition 2008–Graz data set A, Institute for Knowledge Discovery (Laboratory of Brain-Computer Interfaces), Graz University of Technology, 2008.

[16] A. Hyvärinen, E. Oja, Independent component analysis: algorithms and applications, *Neural networks*, 2000, 13(4): 411-430.

[17] A. Hyvarinen, J. Karhunen, E. Oja, Independent Component Analysis, 2007:137-141.

[18] Y. Wang, Y. T. Wang, T. P. Jung, Translation of EEG spatial filters from resting to motor imagery using independent component analysis, *PloS one*, 2012, 7(5): e37665.

[19] H. Ramoser, J. Muller-Gerking, G. Pfurtscheller, Optimal spatial filtering of single trial EEG during imagined hand movement, *Rehabilitation Engineering*, *IEEE Transactions on*, 2000, 8(4): 441-446.

[20] K. Fukunaga. Introduction to statistical pattern recognition, Access Online via Elsevier, 1990.

[21] B. H. Yang, G. Z. Yan, B. Yan, Extracting EEG Feature in Brain-computer Interface Based on Discrete Wavelet Transform, *Chinese Journal of Biomedical Engineering*, 2006, 25(5): 518-521.

[22] WANG Lu, WU Xiao-pei, GAO Xiang-ping, Analysis and Classification of Four-Class Motor Imagery EEG Data, *Computer Technology and Development*, 2008, 18(10): 23-26.

[23] H. Sun, Y. Xiang, Y. Sun, H. Zhu, On-line EEG classification for brain-computer interface based on CSP and SVM, *Image and Signal Processing (CISP)*, *2010 3rd International Congress on*. *IEEE*, 2010, 9: 4105-4108.

[24] G. Pfurtscheller, Final results of BCI Competition IV. Available: http://www.bbci.de/competition/iv/results/#winners